

HARSHIT SINGH SHEKHAWAT

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EDUCATION

Vellore Institute of Technology Bachelor of Technology Major in Computer Science, Specialization in Artificial Intelligence & Machine Learning Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, AWS Cloud Computing, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Computer Architecture	Aug 2023 - Present CGPA: 8.96/10
Delhi Public School Jodhpur Completed secondary and senior secondary education under the CBSE board.	Feb 2009 - May 2023

SKILLS

- Languages: Python, C++, Java, JavaScript, SQL, C
- AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Scikit-learn, TensorFlow, PyTorch
- Web & Backend: Flask, Node.js, Express.js, React.js, REST APIs, HTML, CSS, Bootstrap, Tailwind CSS
- IoT & Embedded: Arduino, ESP32, IoT Sensors
- Databases & Tools: MongoDB, MySQL, Git, Docker, Linux, AWS, MATLAB, R Studio

PROJECTS

<u>Agentic AI Intrusion Detection & Response Pipeline TensorFlow, PyTorch, FastAPI</u> - Engineered an explainable AI-driven IDS using CNN/LSTM/GRU/Transformer models on ~2.8M CICIDS2017 records with graph-based attack correlation, LLM reasoning, and adaptive risk fusion. - Achieved 98.40% accuracy and 99.16% memory retrieval accuracy with human-in-the-loop feedback and retraining workflows.
<u>Smart Parking Analytics & ANPR System OpenCV, YOLOv8, FastAPI, React</u> - Built a computer vision parking system using YOLOv8, EasyOCR, and scikit-learn for real-time occupancy detection and automatic number plate recognition across 396 parking slots. - Developed a FastAPI + React dashboard with live parking analytics, slot recommendation, duplicate suppression, and SQLite-based vehicle tracking.
<u>Smart Irrigation System (AquaField) Flask, Scikit-learn, ESP32, Arduino</u> - Co-developed an ML-powered smart irrigation platform using Flask, ESP32, and IoT sensors to automate irrigation from real-time soil, rain, temperature, and flow data. - Trained and benchmarked 4 regression models on 100K agricultural records, achieving 0.99870 test R² and 328.02 L/ha RMSE with Gradient Boosting. - Implemented real-time dashboard monitoring, rain-aware motor control, and hardware safety cutoffs for efficient irrigation automation.
<u>PadharooGo – Full-Stack Travel Stay Platform Node.js, Express.js, MongoDB, EJS</u> - Engineered a full-stack travel accommodation platform with secure authentication, property listing management, cloud image uploads, and geolocation-based stay discovery. - Integrated Cloudinary, Mapbox, Passport.js, and MongoDB-backed session storage while designing scalable MVC architecture, REST APIs, and complete CRUD workflows.

VOLUNTEER EXPERIENCE

General Secretary – Mharo Rajasthan Club Directed overall club operations by coordinating cultural and technical events, increasing the number of club-conducted events by 40% and helping regain club popularity and student engagement. Led the planning and execution of a large-scale event for 300+ students with overall collections nearing ₹1 lakh while managing coordination, budgeting, and operational workflows.	September 2024 - April 2026
Management Team Member – Linux Club Organized workshops and hackathons focused on Linux, open-source tools, and system-level programming concepts. Managed event planning, scheduling, and team coordination to ensure smooth execution of technical sessions and promote hands-on learning within the student community.	June 2024 - April 2025
Photographer – Photography Club Planned and covered photography events and exhibitions while contributing to creative promotion and visual storytelling activities. Collaborated with club members to improve event outreach, audience engagement, and participation through curated media content.	July 2024 - May 2025